

GDPS/MGM GEOPARM and GEOGROUP



Agenda and objectives



- The purpose of this module is to provide information on the following:
 - Definitions in TOPOLOGY
 - Changes required for both GEOGROUP and GEOPARM for GDPS MGM 4-site configurations
 - How to implement changes to the disk configuration

- Information is presented in the following sequence:
 - General considerations
 - Utility device considerations/recommendations
 - GDPS MGM GEOGROUP changes
 - GDPS MGM GEOPARM changes
 - Implementing changes to the disk configuration

GDPS MGM Utility Devices



- Both GDPS Metro and GDPS Global – GM require the definition of utility devices to identify a target device for issuing certain commands
- PPRC utility device requirements are same as standard GDPS Metro
- GM utility device requirements are basically the same as GDPS GM except you need one for Site1 and one for Site2
- We recommend you reserve one device address per LSS to be used for use as common utility devices (the GDPS Metro utility device)
 - The same device can be used by GEOPARM specifications for the utility PPRC and GM utility devices
- The utility devices must appear in the OSCONFIGs being used by the GDPS systems using them
- If using a common utility device, these devices are recommended to be defined as OFFLINE Yes and SHARED No in the OSCONFIG in HCD to meet the GDPS Metro non-share devices
- You do not have to use the same devices for Metro and GM utility devices, however, doing so keeps things as simple as possible!

OPTIONS - TOPOLOGY



Topology

- In GDPS OPTIONS XML File definitions TOPOLOGY is specified
 - TOPOLOGY = 'MGM4SITE' for a 4-site configuration
 - MGM4SITE = 'MTFO(YES)'
- TOPOLOGY must be the same in all systems in the GDPS sysplex
 - All application systems
 - Both Kp systems
 - Kg system
 - Kr system

GEOGROUP Considerations



GEOGROUP defines pertinent information for GDPS and is mandatory;

- Network Definitions;
Information that enables communication between GDPS Metro and Global Controlling systems;
 - You must define connection definitions between Kp1, Kp2, Kg and Kr
- The Site Table;
Defines all the systems that are part of the GDPS MGM configuration;
 - The Kg-sys Controlling system
 - The Kr-sys Controlling system
 - The Kp-sys Controlling system(s)
 - The Metro Mirror Production and Coupling Facilities
- Alias and Partner Definitions;
Relationship definitions between Controlling systems are used in Remote Script Execution (RSE)
- GEOGROUP must be the same in all GDPS Controlling systems
- An alternate GEOGROUP should be prepared for moving the Kg-sys instance between systems

GEOGROUP Considerations

Refreshing the Site Table;

To update the Site Table elements of GEOGROUP (Location, System and Domain elements) will require a recycle of ALL GDPS Netviews starting with the Controlling System(s).



GEOGROUP Considerations - Network



- GEOGROUP holds information that enables all K-systems to communicate with each other.
- You must define connection definitions between Kp1, Kp2, Kg and Kr

```
<configuration>
  <Network Name="MGM4">
    <Connection Alias="KPA1" Netid="NETWRKA" IPstack="TCPIP"
      IPAddress="9.212.128.139"
      LowPriorityPort="4022"
      libertyProtocol="https" libertyPort="9080"
      daemonPort="1995">
    </Connection>
    <Connection Alias="KPA2" Netid="NETWRKA" IPstack="TCPIP"
      IPAddress="9.212.128.149"
      LowPriorityPort="4022"
      libertyProtocol="https" libertyPort="9080"
      daemonPort="1995">
    </Connection>
    <Connection Alias="KGAB" Netid="NETWRKA" IPstack="TCPIP"
      IPAddress="9.212.128.149"
      LowPriorityPort="4024"
      libertyProtocol="https" libertyPort="9080"
      daemonPort="1995">
    </Connection>
    <Connection Alias="KRBA" Netid="NETWRKA" IPstack="TCPIP"
      IPAddress="9.212.128.149"
      LowPriorityPort="4026"
      libertyProtocol="https" libertyPort="9080"
      daemonPort="1995">
    </Connection>
```

GEOGROUP Considerations - Network



- GEOGROUP holds information that enables all K-systems to communicate with each other.
- You must define connection definitions between Kp1, Kp2, Kg and Kr

```
<Connection Alias="KPB1" Netid="NETWRKB" IPstack="TCPIP"
            IPaddress="9.212.128.113"
            LowPriorityPort="4022"
            libertyProtocol="https" libertyPort="9080"
            daemonPort="1995">
</Connection>
<Connection Alias="KPA2" Netid="NETWRKB" IPstack="TCPIP"
            IPaddress="9.212.128.114"
            LowPriorityPort="4022"
            libertyProtocol="https" libertyPort="9080"
            daemonPort="1995">
</Connection>
<Connection Alias="KGAB" Netid="NETWRKB" IPstack="TCPIP"
            IPaddress="9.212.128.113"
            LowPriorityPort="4024"
            libertyProtocol="https" libertyPort="9080"
            daemonPort="1995">
</Connection>
<Connection Alias="KRAB" Netid="NETWRKB" IPstack="TCPIP"
            IPaddress="9.212.128.113"
            LowPriorityPort="4022"
            libertyProtocol="https" libertyPort="9080"
            daemonPort="1995">
</Connection>
</Network>
```


GEOGROUP Considerations – Region A Site-Table

The names of the Regions and Replication sites are defined in GEOGROUP



```
<Group Name="MGMGRP" Type="MGM4SITE">
<Region Name="NEW_YORK_CITY" NickName="NYC" Location="Local">
<ServerSite Name="Campus1" NickName="SS1" Location="Site1">
<Location Name="lpar1" Lpar="NYC.GKA1L"></Location>
<Location Name="lpar2" Lpar="NYC.PR11L"></Location>
<Location Name="lpar3" Lpar="NYC.CFA1L"></Location>
<System Name="GKA1" Type="zOS" Location="lpar1">
  <Domain Name="KPA1" Type="KP"
    Alias="KPA1" Sysplex="A0GDPS"></Domain>
  <Domain Name="KGA1" Type="KG"
    Alias="KGA1" Sysplex="A0GDPS"></Domain>
  <Domain Name="KRA1" Type="KG"
    Alias="KRA1" Sysplex="A0GDPS"></Domain>
</System>
<System Name="PRD1" Type="zOS" Location="lpar2">
  <Domain Name="NVP1" Type="PROD"></Domain>
</System>
<System Name="CFA1" Type="CF" Location="lpar3">
</System>
</ServerSite>
<ServerSite Name="Campus2" NickName="SS2" Location="Site2">
<Location Name="lpar4" Lpar="NYC.GKA2L"></Location>
<Location Name="lpar5" Lpar="NYC.PR21L"></Location>
<Location Name="lpar6" Lpar="NYC.CFA2L"></Location>
<System Name="GKA2" Type="zOS" Location="lpar4">
  <Domain Name="KPA2" Type="KP"
    Alias="KPA2" Sysplex="A0GDPS"></Domain>
</System>
<System Name="PRD2" Type="zOS" Location="lpar5">
  <Domain Name="NVP2" Type="PROD"></Domain>
</System>
<System Name="CFA2" Type="CF" Location="lpar6">
</System>
</ServerSite>
</Region>
```

Location element

The Location element is mandatory for all LPARs in the ServerSite:

```
<Location Name="lpar2" Lpar="NYC.PR11L"></Location>
```

System element

The System element is mandatory for all systems in the ServerSite.

The order of the System element within a ServerSite is reflected in the GDPS Standard Action Panels:

```
<System Name="PRD1" Type="zOS" Location="lpar2">
.....
</System>
```

Domain element

The Domain element is mandatory for all systems defined as type zOS in the system element. The Sysplex and Alias attributes are mandatory for systems that have their alias specified in the Connection element. Multiple Domain elements can be specified for a system where more than one Netview is running:

```
<System Name="GKA1" Type="zOS" Location="lpar1">
<Domain Name="KPA1" Type="KP" Alias="KPA1" Sysplex="A0GDPS"></Domain>
<Domain Name="KGA1" Type="KG" Alias="KGA1" Sysplex="A0GDPS"></Domain>
</System>
```

GEOGROUP Considerations – Region B Site-Table



The names of the Regions and Replication sites are defined in GEOGROUP

```
<Region Name="London" NickName="LON" Location="Remote">
<ServerSite Name="Campus3" NickName="SS1" Location="SITE1">
<Location Name="lpar1" Lpar="LON.GKB1L"></Location>
<Location Name="lpar2" Lpar="LON.PR11L"></Location>
<Location Name="lpar3" Lpar="LON.CFB1L"></Location>
<System Name="DRKB" Type="zOS" Location="lpar1">
  <Domain Name="KPB1" Type="KP"
    Alias="KPB1" Sysplex="A0GDPS,B0GDPS"></Domain>
  <Domain Name="KGB1" Type="KG"
    Alias="KGB1" Sysplex="A0GDPS,B0GDPS"></Domain>
  <Domain Name="KRB1" Type="KR"
    Alias="KRB1" Sysplex="A0GDPS,B0GDPS"></Domain>
</System>
<System Name="PRD1" Type="zOS" Location="lpar2">
  <Domain Name="NVP1" Type="PROD"></Domain>
</System>
<System Name="CFB1" Type="CF" Location="lpar3">
</System>
</ServerSite>

<Region Name="London" NickName="LON" Location="Remote">
<ServerSite Name="Campus4" NickName="SS1" Location="SITE1">
<Location Name="lpar4" Lpar="LON.DRKBL"></Location>
<Location Name="lpar5" Lpar="LON.PR11L"></Location>
<Location Name="lpar6" Lpar="LON.CFB1L"></Location>
<System Name="GKB2" Type="zOS" Location="lpar4">
  <Domain Name="KPB2" Type="KP"
    Alias="KPB2" Sysplex="A0GDPS,B0GDPS"></Domain>
</System>
<System Name="PRD2" Type="zOS" Location="lpar5">
  <Domain Name="NVP2" Type="PROD"></Domain>
</System>
<System Name="CFB2" Type="CF" Location="lpar6">
</System>
</ServerSite>
</Region>
```

Location element

The Location element is mandatory for all LPARs in the ServerSite:

```
<Location Name="lpar2" Lpar="LON.PR11L"></Location>
```

System element

The System element is mandatory for all systems in the ServerSite.

The order of the System element within a ServerSite is reflected in the GDPS Standard Action Panels:

```
<System Name="PRD1" Type="NOGDPS" Location="lpar2">
```

```
.....
</System>
```

Domain element

The Domain element is mandatory for all systems defined as type zOS in the system element. The Sysplex and Alias attributes are mandatory for systems that have their alias specified in the Connection element. Multiple Domain elements can be specified for a system where more than one Netview is running:

```
<Domain Name="KRB1" Type="KR" Alias="KRB1" Sysplex="B0GDPS"></Domain>
```

GEOGROUP Considerations – AliasGroup



The relationships between the K-systems are described in the Alias and Partner definitions

```
<AliasGroup
    Name="KPAA"
    Domains="KPA1 KPA2">
</AliasGroup>
<AliasGroup
    Name="KPBB"
    Domains="KPB1 KPB2">
</AliasGroup>
</Group>
```

The AliasGroup defines the systems that can be addressed. Which system will be addressed is determined at runtime

If one of the Kp-systems are addressed (KpA1 or KpA2) GDPS will at execution time determine which one is the Master System.

GEOPARM Considerations



- GDPS Metro and GDPS GM will have the disk configurations described in a single GEOPARM file definition.
- The devices defined in the GEOPARM do not have to be 1:1 defined in both replication legs
 - Metro Mirror can have more devices than those in GM
 - GM can have more devices than those defined in Metro Mirror
 - The additional devices in GM are not supported for IR configurations
 - Care must be taken to ensure your config meets your HA/DR requirements
- MSS1 is supported in GDPS/MGM 4-site configurations as follows:
 - GM FlashCopy devices (or these also support no UCB definition)
 - MM RS2 devices in Region B
 - For MGM IR configurations only
 - Non-IR configurations can be tolerated where Kg sys uses a different IODF or channel paths.
- FB devices are not supported for MGM IR configurations using the IR Tool or the GDPS supplied Procedures

GDPS GEOPARM major changes



- One consolidated GEOPARM will define both the Metro Mirror leg and the Global Mirror leg
- In GEOPARM definitions for all three sites must be described
- The Logical Subsystem Group definitions must be defined for both the Metro Mirror – and the GM leg
- One possible change you might want to make is if you have a non-IR configuration and choose to run the Kg-sys in one of your GDPS application systems, you will want to include any additional disk for the Kg-sys into the GEOPARM.
- It is always recommended to run the Kg system in the opposite site from the site that drives the Global Mirror
 - GM running cascaded from RS2 in site 2 ➔ Run Kg in Site 1
 - GM running Multi Target from RS1 in site 1 ➔ Run Kg in Site 2

GEOPARM Region and Replication Site definitions



```
<DiskConfiguration>
<Region Name="NEW_YORK">
  <ReplicationSite Name="NYC1" NickName="NY1"
    Location="Site1">
    <DiskSubsystem Name="NY11" Serial="00FGLP1"
      UtilityDevice0="00.00" </DiskSubsystem>
    <DiskSubsystem Name="NY21" Serial="00FGLP2"
      UtilityDevice0="00.00" </DiskSubsystem>
  </ReplicationSite>
  <ReplicationSite Name="NYC2" NickName="NY2"
    Location="Site2">
    <DiskSubsystem Name="NY21" Serial="00FGLP3"
      UtilityDevice0="00.00" </DiskSubsystem>
    <DiskSubsystem Name="NY22" Serial="00FGLP4"
      UtilityDevice0="00.00" </DiskSubsystem>
  </ReplicationSite>
</Region Name="LONDON">
  <ReplicationSite Name="LON1" NickName="LN1"
    Location="Site1">
    <DiskSubsystem Name="LN11" Serial="00MV331"
      UtilityDevice0="00.00" </DiskSubsystem>
    <DiskSubsystem Name="LN12" Serial="00MV332"
      UtilityDevice0="00.00" </DiskSubsystem>
  </ReplicationSite>
  <ReplicationSite Name="LON2" NickName="LN2"
    Location="Site2">
    <DiskSubsystem Name="LN21" Serial="00MV333"
      UtilityDevice0="00.00" </DiskSubsystem>
    <DiskSubsystem Name="LN22" Serial="00MV334"
      UtilityDevice0="00.00" </DiskSubsystem>
  </ReplicationSite>
</Region>
```

- Region definitions must be made
- Four replication sites defined
 - Two in the metro region
 - Two in the DR region
- A Utilitydevice0 or a Utilitydevice1 statement will point to a device that is a utility device for the complete disk system. LSS.CCA format is used
 - Utilitydevice0 requires an even LSS
 - Utilitydevice1 requires an odd LSS
 - (device in cluster0 or cluster1)
- Location MUST be SITE1 or SITE2.
- Other parameters selected by customer
- Names specified must match the names specified in GEOGROUP

GEOPARM Consistency Group and Session Master



```
<ConsistencyGroup Name="GMPROD">  
  SNBR="2A,2B"  
  CGDRAIN="0"  
  CGINTERVAL="0"  
  COORDINTERVAL="0">  
  
<SessionMaster>AS11,AS21,BS11,BS21</SessionMaster>
```

The **ConsistencyGroup** element defines the Global Mirror session that is in the configuration.

The **SessionMaster** element lists the disk subsystems which can serve as the GM session Master when GM is running in the replication site where the disk subsystem id located

SNBR specifies the Global Mirror Session Number. In a Three-site configuration you only specify one SNBR. In a Four-site configuration you specify a number for the session running from Region A to Region B and a number for the opposite direction. It is recommended to specify two different numbers.

SessionSubordinate definitions are used where you need to define multiple DSS in each site for the GM topology, no different to GM 2-site.

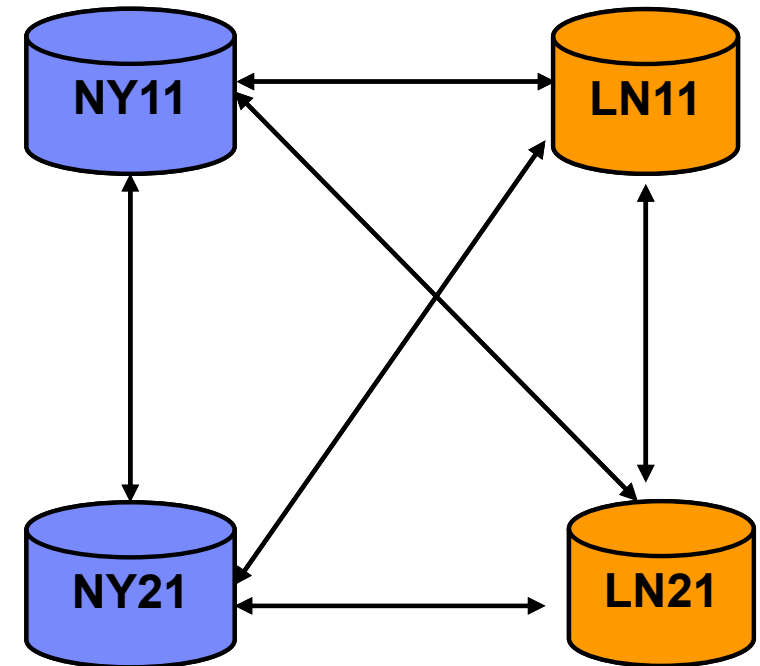
GEOARM disk definition



```
<MemberDiskSubsystem>RegionA_RS1,RegionA_RS2_DSS,RegionB_RS1_DSS,RegionB_RS2_DSS
</MemberDiskSubsystem>
```

- Four sets of disks must be defined in the MemberDiskSubsystem definitions within the LogicalSubsystemGroup element

```
<MemberDiskSubsystem>NY11,NY21, LN11, LN21
</MemberDiskSubsystem>
<Path FromDSS="NY11"
      ToDSS="NY21">
  <Link>01000100</Link>
</Path>
<Path FromDSS="NY21"
      ToDSS="LN11">
  <Link>03000300</Link>
</Path>
<Path FromDSS="NY11"
      ToDSS="LN11">
  <Link>02000200</Link>
</Path>
<Path FromDSS="LN11"
      ToDSS="LN21">
  <Link>01000100</Link>
</Path>
<Path FromDSS="NY21"
      ToDSS="LN21">
  <Link>02000200</Link>
</Path>
<Path FromDSS="LN21"
      ToDSS="NY11">
  <Link>03000300</Link>
</Path>
```



- Link definitions between all set of disks must be defined

GEOPARM Logical Subsystem Definitions



```
<LogicalSubsystem>
<LSS>00,00,00,00</LSS>
<GMFCLSS>70,,70,</GMFCLSS>
<UtilityDevice>00,00,00,00</UtilityDevice>
<DeviceSet>
  <Mirror CopyOnce="YES">01,01,01,01</Mirror>
  <GMFC>01,,01,</GMFC>
  </Count>3</Count>
</DeviceSet>
<DeviceSet>
  <Mirror>04,04,04,04</Mirror>
  <GMFC>04,,04,</GMFC>
  </Count>28</Count>
</DeviceSet>
```

LSS: Defines the LSSs which contain the mirrored devices. The CCAs of the mirrored devices are defined in the Mirror element found in the DeviceSet statement

GMFCLSS: Defines the LSSs which contain the Flashcopy devices for GM. Only necessary if different from LSS associated with the mirrored devices. CCAs are defined in the GMFC statement

FC1LSS: Defines the LSSs containing the Flashcopy devices assigned to practice. The corresponding CCAs will be defined in the FC1 statement. This statement is optional.

UtilityDevice: Define one Utility Device for each LSS

Mirror: Defines a range of consecutive devices that will be mirrored. Can be specified as a range or a start address followed by a count statement
Copy Once devices also defined here

Example: <Mirror>01-03,01-03,01-03,01-03</Mirror> is the same as
<Mirror>01,01,01,01</Mirror>
<Count>03</Count>